

SAMANTHA REED - RESUME

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PROFESSIONAL SUMMARY

Hardware & Embedded Systems Engineer with strong expertise in microcontrollers, signal processing, PCB design, and firmware development. Passionate about IoT devices, VLSI systems, and embedded software integration.

TECHNICAL SKILLS

- Hardware & ECE: Embedded Systems, Microcontrollers, PCB Design, VLSI, Signal Processing, Electronics
- Software & Tools: C++, Python, Git, Data Structures, Communication Systems

PROFESSIONAL EXPERIENCE

Embedded Systems Engineer | MicroWave Tech (2022 - Present)

- Designed custom 4-layer Printed Circuit Boards (PCB Design) for IoT sensor nodes using microcontrollers.
- Developed real-time firmware in C++ and Python for sensor data acquisition and signal processing.
- Optimized communication systems protocols (I2C, SPI, UART) between peripheral hardware sensors.
- Implemented VLSI chip layout verifications and digital logic design validation.

Electronics Hardware Engineer | Circuit Dynamics (2020 - 2022)

- Tested and debugged high-frequency analog and digital electronics components.
- Applied signal processing algorithms to filter environmental sensor noise in embedded systems.

EDUCATION

Bachelor of Engineering in Electronics & Communication (ECE) | Tech Institute of Engineering (2016 - 2020)

- Specialization: Embedded Systems, VLSI, and Signal Processing.